

Topological Connected Component Partitioning for One-Class Null Foley-Sammon Transform

1 Introduction

In one-class classification, the goal is to characterize the distribution of nominal data to detect outliers. Standard Null Foley-Sammon Transform (NFST) approaches often rely on heuristic partitioning, such as K -means, to generate pseudo-labels. However, these methods assume hyperspherical clusters and fail to capture complex, non-convex manifold structures. We propose a ****Topological Manifold Partitioning (TMP)**** approach that discovers the natural connectivity of the data using graph-based contraction, extracting stable connected components as the basis for the NFST subspace.

2 Topological Manifold Partitioning (TMP)

Instead of selecting anchors or cluster centers, we build a global representation of the data and decompose it into topologically consistent components.

2.1 Graph Construction and Lifetime Calculation

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$ be the training set. We construct a k -nearest neighbor (k -NN) graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where an edge exists between $\mathbf{x}_i$ and $\mathbf{x}_j$ if they are among each other's k closest neighbors.

To identify the "core" structure of the manifold and eliminate density bridges, we apply an iterative graph contraction process. Let $r_i^{(t)}$ be the degree (rank) of node i at iteration t . In each step, nodes with the minimum current rank are removed:

$$\tau_i = t \quad \text{s.t.} \quad r_i^{(t)} = \min_{j \in \mathcal{V}_{\text{alive}}} r_j^{(t)}. \quad (1)$$

The resulting *Lifetime* τ_i represents the structural persistence of each sample. Samples with high τ values are located in the dense, central skeletons of the manifold, while low τ values indicate boundary points or noise.

2.2 Spectral-based Component Partitioning

Once the weight matrix $\mathbf{W}$ of the k -NN graph is constructed, we can extract its manifold structure through its spectral properties. Instead of manual label propagation, we utilize the eigenvectors of the Graph Laplacian to identify the connected components (or weakly connected dense regions).

1. **Graph Laplacian Construction:** Compute the degree matrix $\mathbf{D}$ where $D_{ii} = \sum_j W_{ij}$. The normalized random-walk Laplacian is defined as:

$$\mathbf{L}_{rw} = \mathbf{I} - \mathbf{D}^{-1}\mathbf{W} \quad (2)$$

2. **Eigen-decomposition:** Solve the eigenvalue problem $\mathbf{L}_{rw}\mathbf{v} = \lambda\mathbf{v}$. According to spectral graph theory, the number of eigenvalues $\lambda = 0$ corresponds to the number of connected components in the graph.

3. **Manifold Embedding:** We select the c eigenvectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_c]$ corresponding to the c smallest eigenvalues. Each row of $\mathbf{V}$ provides a low-dimensional spectral embedding of the original data points $\mathbf{x}_i$.
4. **Pseudo-label Assignment:** By applying a simple clustering (e.g., K -means) on the rows of $\mathbf{V}$, we obtain the pseudo-labels $\mathbf{y} = \{y_1, \dots, y_n\}$. These labels identify the distinct topological components of the manifold.

This spectral approach is more robust than simple connectivity, as the eigenvectors can effectively "cut" weak density bridges that would otherwise merge distinct manifold structures.

3 One-Class NFST via Spectral Components

With the discovered connected components, we apply the Null Foley-Sammon Transform to find a subspace where within-component variance is minimized and between-component variance is maximized.

3.1 Topological Scatter Matrices

The within-class scatter matrix $\mathbf{S}_w$ and the between-class scatter matrix $\mathbf{S}_b$ are computed using the topological pseudo-labels:

$$\mathbf{S}_w = \frac{1}{n} \sum_{j=1}^c \sum_{\mathbf{x}_i \in C_j} (\mathbf{x}_i - \boldsymbol{\mu}_j)(\mathbf{x}_i - \boldsymbol{\mu}_j)^T, \quad \mathbf{S}_b = \frac{1}{n} \sum_{j=1}^c n_j (\boldsymbol{\mu}_j - \boldsymbol{\mu})(\boldsymbol{\mu}_j - \boldsymbol{\mu})^T, \quad (3)$$

where $\boldsymbol{\mu}_j$ is the mean of the j -th connected component and $\boldsymbol{\mu}$ is the global mean.

3.2 Null Space Transformation

We seek a projection matrix $\boldsymbol{\Theta}$ that satisfies the NFST criterion:

$$\boldsymbol{\Theta}^T \mathbf{S}_w \boldsymbol{\Theta} = 0 \quad \text{and} \quad \boldsymbol{\Theta}^T \mathbf{S}_b \boldsymbol{\Theta} > 0. \quad (4)$$

The transformation $\boldsymbol{\Theta}$ maps all nominal samples belonging to the same topological structure to a single point in the projected space.

3.3 Anomaly Scoring

For a test sample $\mathbf{x}^*$, the anomaly score is defined as the distance to the nearest topological component in the Null Space:

$$\mathcal{A}(\mathbf{x}^*) = \min_{j \in \{1, \dots, c\}} \|\boldsymbol{\Theta}^T \mathbf{x}^* - \boldsymbol{\Theta}^T \boldsymbol{\mu}_j\|_2^2. \quad (5)$$

After obtaining the pseudo-labels $\mathbf{y}$ from the spectral partitioning, we compute the scatter matrices $\mathbf{S}_w$ and $\mathbf{S}_b$ as defined in Eq. (2-3). The NFST then solves for the projection matrix $\boldsymbol{\Theta}$ that lies in the null space of $\mathbf{S}_w$.

Algorithm 1 Spectral-Partitioned One-Class NFST

- 1: **Input:** Training set $\mathbf{X}$, number of components c .
 - 2: **Step 1: Spectral Discovery**
 - 3: - Construct the adjacency matrix $\mathbf{W}$ (e.g., using a Gaussian kernel on k -NN).
 - 4: - Compute the Normalized Laplacian $\mathbf{L}_{rw} = \mathbf{I} - \mathbf{D}^{-1}\mathbf{W}$.
 - 5: - Extract the c eigenvectors $\mathbf{V}$ associated with the smallest eigenvalues.
 - 6: - Assign pseudo-labels y_i by clustering the rows of $\mathbf{V}$.
 - 7: **Step 2: NFST Subspace Learning**
 - 8: - Compute $\mathbf{S}_w$ and $\mathbf{S}_b$ based on the spectral pseudo-labels y_i .
 - 9: - Solve the generalized eigenvalue problem to find the Null Space Θ .
 - 10: **Step 3: Scoring**
 - 11: - For a test sample $\mathbf{x}^*$, project it: $\mathbf{z}^* = \Theta^T \mathbf{x}^*$.
 - 12: - Calculate $\mathcal{A}(\mathbf{x}^*) = \min_j \|\mathbf{z}^* - \text{centroid}(\mathbf{z}_{C_j})\|^2$.
 - 13: **return** Anomaly score $\mathcal{A}(\mathbf{x}^*)$.
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